Assignment 3

This exercise sheet is devoted to the LU decomposition. It is meant to better understand how it works, how it is implemented and what is the role of pivoting.

Graded exercise

Exercise 13 (LU-decomposition without pivoting)
Let $A \in \mathbb{R}^{m \times m}$ be an invertible matrix, and let $b \in \mathbb{R}^m$, for $m \in \mathbb{N}$. The goal of this exercise is to implement the LU-decomposition without pivoting and use it to solve a linear system. You are therefore required to use only elementary operations in the implementation. In particular, you cannot use the PYTHON function `scipy.linalg.lu` or other built-in linear algebra functions unless explicitly specified.

Note that it might be easier for your understanding to first solve task (c), which requires handwritten calculations, and then move to the programming tasks (a), (b) and (d).

(a) Implement a function `lu_decomposition` that, given a matrix A in input, computes its LU-decomposition without pivoting. Instead of allocating new matrices for L and U, the function should return the matrix A that has been overwritten, with L in its lower triangular part and U in its upper triangular part. Note that, for L, the diagonal does not need to be saved.

Hint: For the implementation, you might want to start by writing a code where the matrices L and U are explicitly stored and then modify it to have the entries overwritten in A.

Download the PYTHON files `backward_solve.py` and `forward_solve_mod.py` provided in Brightspace. They contain functions to solve a upper resp. lower triangular system using backward resp. forward substitution. The routine for forward substitution assumes that the matrix has all ones on the diagonal.

(b) Using the function you implemented in (a) and the two you have downloaded, implement a function `lu_solve` that takes in input a matrix A and a right-hand side b, and returns, in an array, the solution v to the linear system $Av = b$ computed using the LU-decomposition.

We now test the routines written in the previous tasks.

(c) Consider the matrix and right-hand side

$$A = \begin{pmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 3 \\ 19 \\ 0 \end{pmatrix}.$$

Compute *by hand* the solution to the linear system $Av = b$, by first computing the LU-decomposition of A, and then performing forward and backward substitution. Do not forget to hand in all your calculations next to the final result.

(d) Write now a PYTHON script that uses the routine from task (b) to compute the solution v to $Av = b$, with A and b as in the previous task. The script should print the result v on screen (which should be equal to the result you obtained by hand in task (c)).

Other exercises

Exercise 14 (Diagonally dominant matrices)
A matrix $A \in \mathbb{R}^{m\times m}$ is called *strictly diagonally dominant* if its entries satisfy:

$$|a_{ii}| > \sum_{j\neq i} |a_{ij}|, \quad \text{for all } i = 1, \dots, m.$$

Prove that, for a strictly diagonally dominant matrix, it is possible to compute its LU-decomposition without pivoting.
Hint: Start by showing that no pivoting is needed for the first row. It is then sufficient to show that also the $(m-1)\times(m-1)$ matrix, with row and column indices $i, j = 2, \dots, m$ resulting after Gaussian elimination for the first row, is strictly diagonally dominant.

Exercise 15 (Pivoting in LU-decomposition)

(a) Consider the following matrix:

$$A = \begin{pmatrix} 4 & 3 & 10 \\ 2 & 4 & 10 \\ 2 & 1 & 3 \end{pmatrix}.$$

Compute (by hand) the LU-decomposition of A using pivoting.

(b) Consider the following matrices:

$$M = \begin{pmatrix} 2 & 4 & 1 \\ 3 & 6 & 2 \\ 0 & 3 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{pmatrix}.$$

For each of these matrices, is pivoting needed? Motivate your answers.

Hint: For the matrix N, the results from the previous exercise may be useful.

(c) Consider a strictly diagonally dominant matrix of the form

$$B = \begin{pmatrix} * & * & * & * & * \\ * & * & & & \\ * & & * & & \\ * & & & * & \\ * & & & & * \end{pmatrix} \in \mathbb{R}^{5\times 5},$$

where $*$ denotes a non-zero entry. Show that, when computing the LU-decomposition (without pivoting), the first step in the Gaussian elimination turns all entries that were originally 0 into non-zero entries.

This phenomenon is known as *fill-in*. It is highly undesirable when using the sparse format to store the matrices, as it leads to a considerable increase in the amount of memory required.

(d) Find two permutation matrices P_1 and P_2 such that, when computing the LU-decomposition of $P_1 B P_2$ (without pivoting), all entries that were originally 0 remain so.